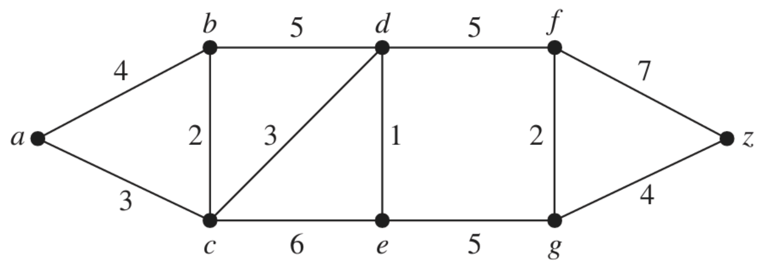
Leading University
Department of Computer Science and Engineering
CSE – 1151 (Discrete Mathematics)
Quiz Exam

Full Marks: 10 (2 x 5) *[Answer any 2 questions]* Time: 20 mins

1. The message "PHHW BRX LQ WKH SDUN" was encrypted using a Caesar cipher. Decrypt the message to find the original text.

2. Using the Fast Modular Exponentiation method, find 8^{224} mod 11.

3. Using Dijkstra's algorithm, find the shortest path and the length between a and z of the following weighted graph.



Leading University
Department of Computer Science and Engineering
CSE – 2231 (Data Communications)
Tutorial Exam

Full Marks: 10 (2 x 5) *[Answer any 2 questions]* Time: 20 mins

1. In a digital transmission system, the receiver clock is 0.15 percent slower than the sender clock. How many more or fewer bits per second does the receiver interpret if the data rate is 1 kbps? How many if the data rate is 1 Mbps?

2. A sampled signal with an amplitude range of -20V to +20V is uniformly quantized into 8 levels. Given the PAM values 3.6, 5.6, 14.9, 19.8, 16.5, 6.5, -1.4, -5.6, and -10.9V, calculate the normalized quantization error for each sample.

3. Draw the digital signal representation of the data stream "10100000100001" using NRZ-L, NRZ-I, Manchester, and differential Manchester encoding schemes.

4. A signal is carrying data in which one data element is encoded as two signal elements. If the bit rate is 100 kbps, what is the average value of the baud rate if c is between 0 and 1?